

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Purchases

Transactional records analyzed

18

Features

Demographics, purchase details & behavior

37

Missing Values

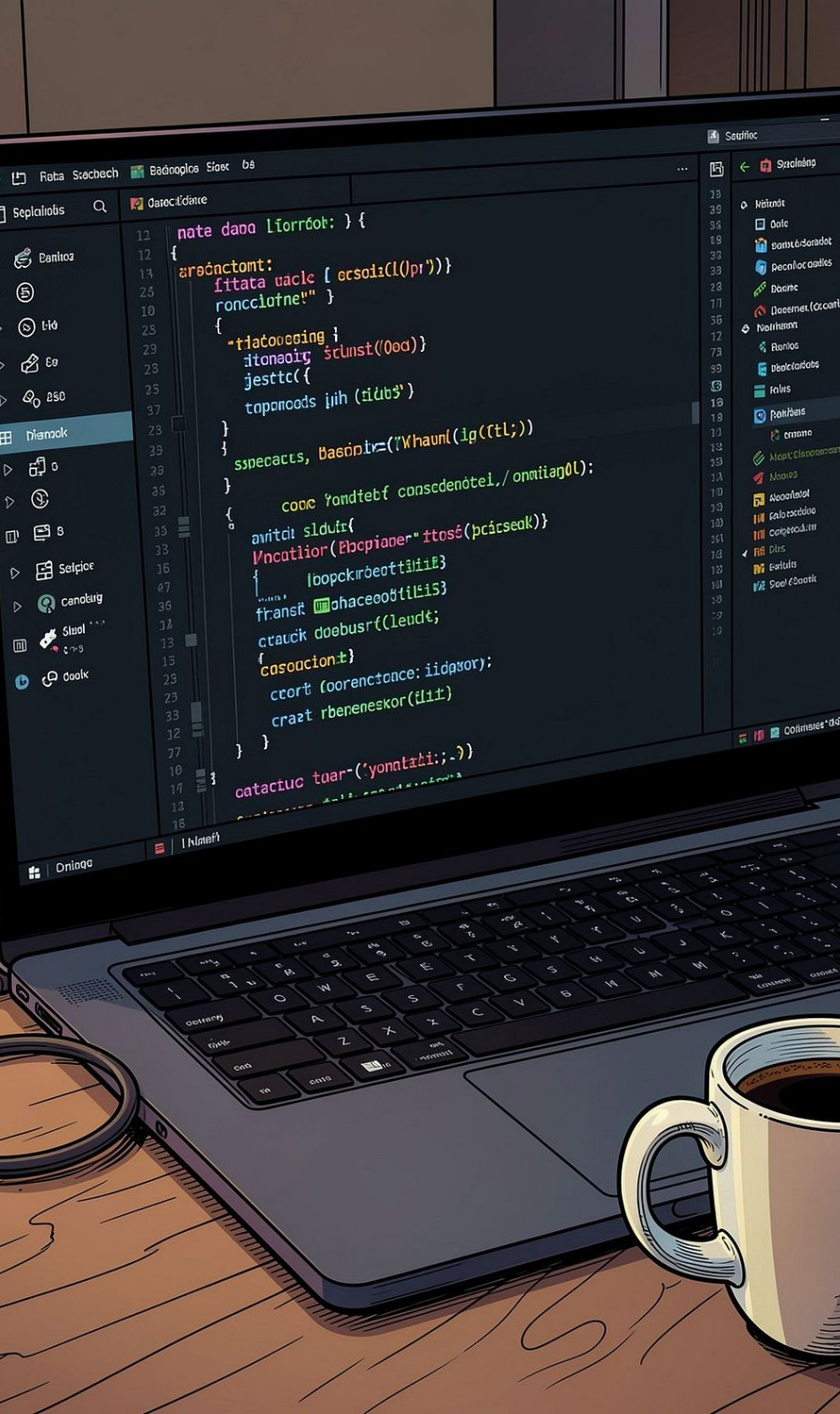
In Review Rating column only

\$59.76

Avg. Purchase

Mean transaction value (USD)

Key features include customer demographics (Age, Gender, Location, Subscription Status), purchase details (Item, Category, Amount, Season, Size, Color), and shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency, Review Rating, Shipping Type).



Data Preparation in Python

1

Load & Explore

Imported via `pandas`; used `df.info()` and `.describe()` for structure and summary stats.

2

Clean Missing Data

Imputed missing Review Ratings using median rating per product category.

3

Feature Engineering

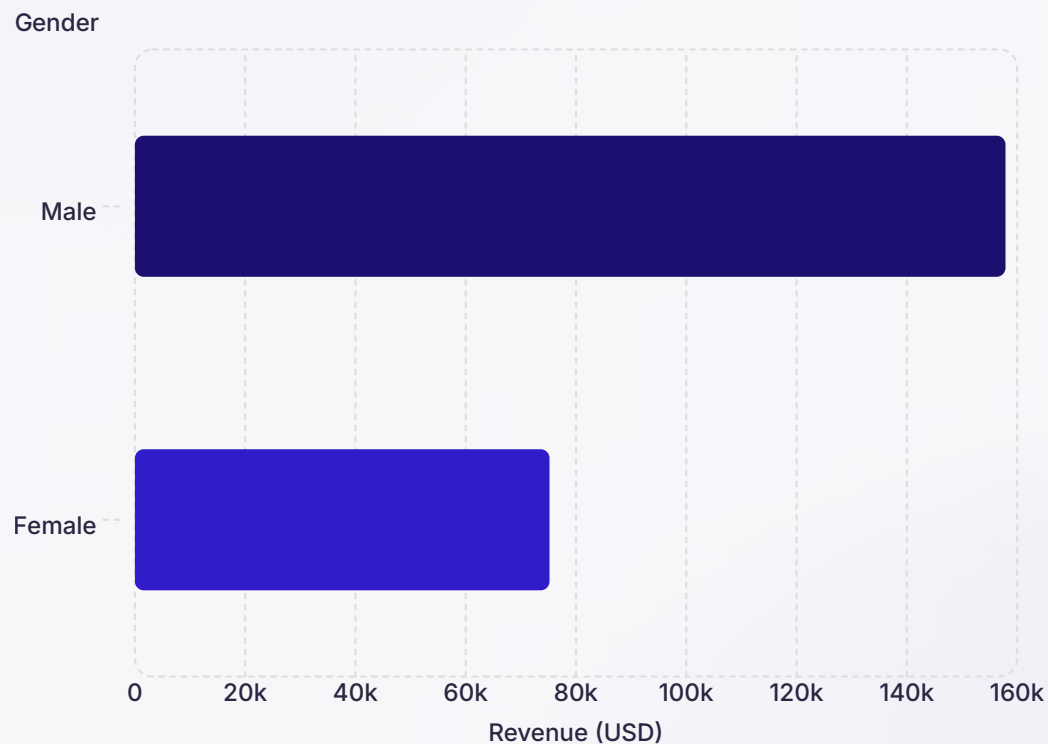
Created `age_group` bins and `purchase_frequency_days`; dropped redundant `promo_code_used`.

4

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Revenue by Gender



Key Finding

Male customers generated **more than double** the revenue of female customers — \$157,890 vs. \$75,191.

This suggests a significant opportunity to better understand and engage the female customer segment.

Subscription & Discount Insights

Subscribers vs. Non-Subscribers

Non-subscribers outnumber subscribers nearly **3:1** (2,847 vs. 1,053), yet average spend is nearly identical — \$59.87 vs. \$59.49.

Subscribers represent a high-potential segment for loyalty incentives.

Most Discount-Dependent Products

Hat — 50.00%

Sneakers — 49.66%

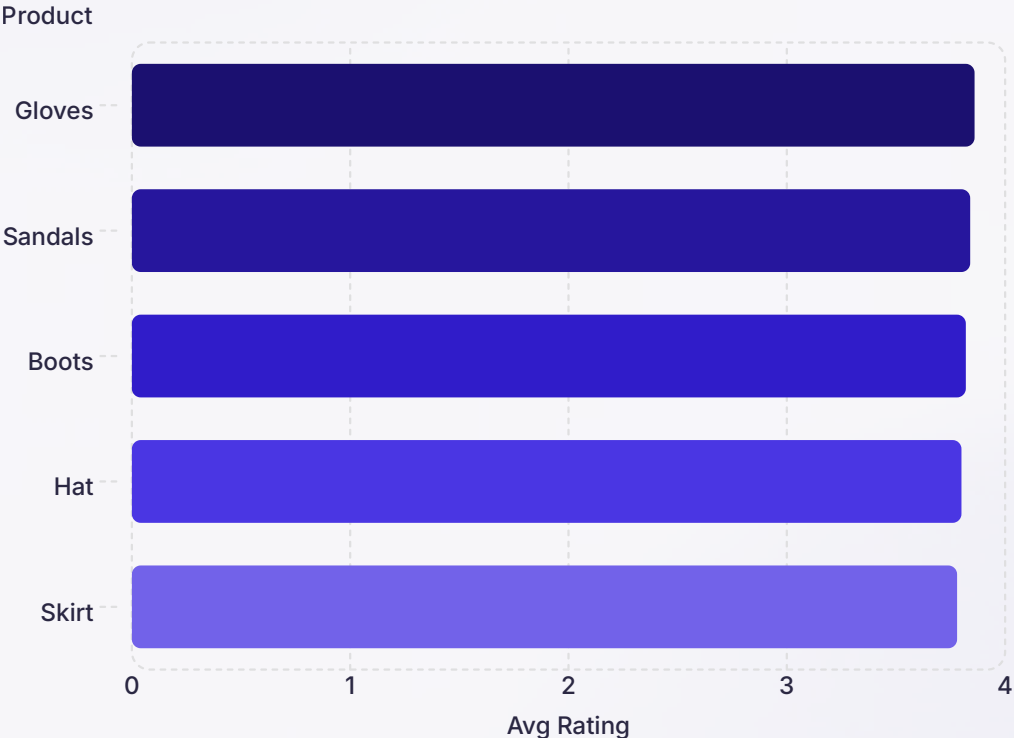
Coat — 49.07%

Sweater — 48.17%

Pants — 47.37%

Top Products by Rating & Category

Top 5 by Average Rating



Top Products per Category

Accessories

Jewelry (171),
Sunglasses (161), Belt
(161)

Clothing

Blouse (171), Pants (171),
Shirt (169)

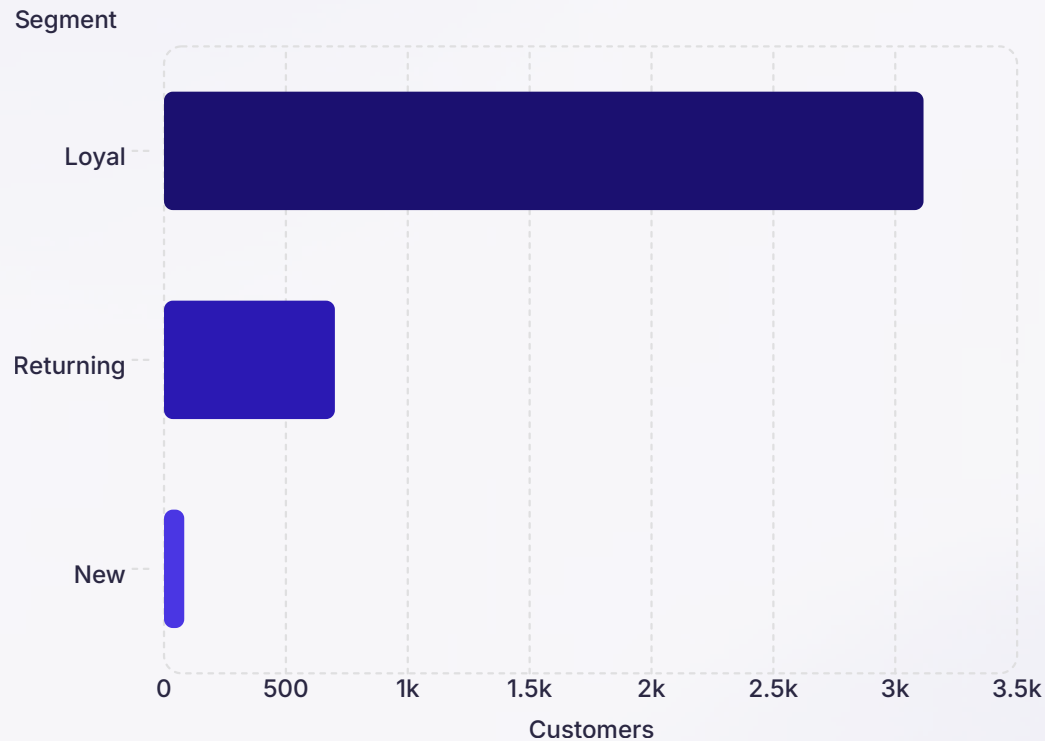
Footwear

Sandals (160), Shoes
(150), Sneakers (145)

Outerwear

Jacket (163), Coat (161)

Customer Segmentation



Segment Breakdown

Customers classified into three segments based on purchase history:

Of customers with more than 5 purchases, **958 are subscribers** vs. **2,518 non-subscribers**, suggesting repeat buyers are not yet converting to subscriptions at scale.

→ **Loyal — 3,116**

80% of the customer base

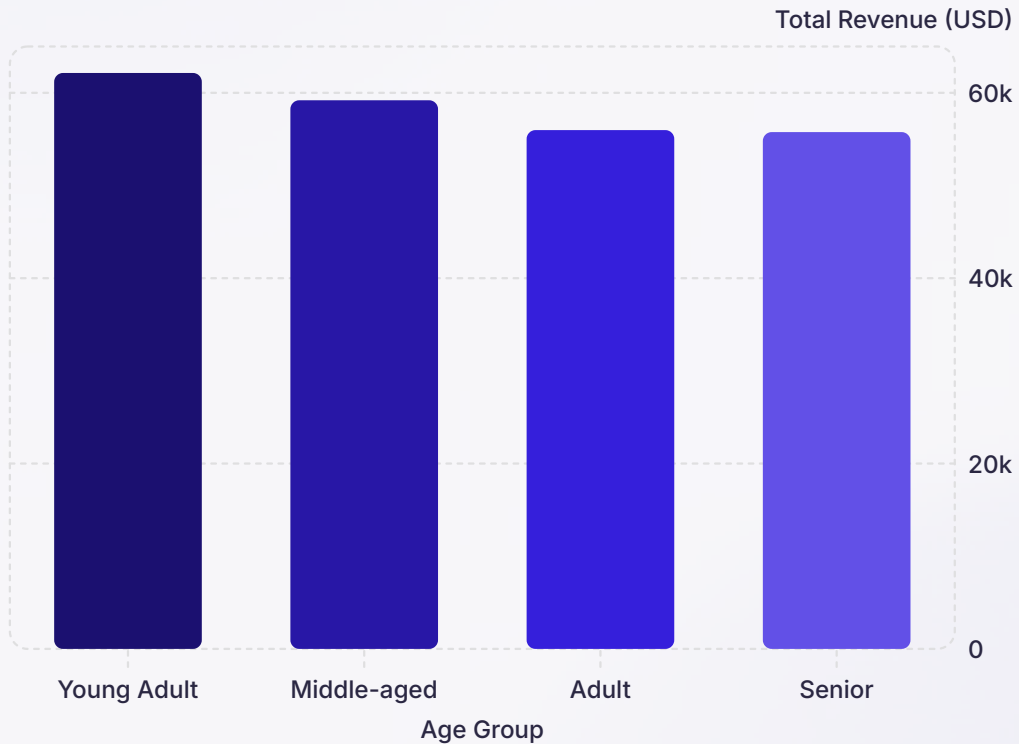
→ **Returning — 701**

18% — potential to convert to Loyal

→ **New — 83**

Only 2% — growth opportunity

Revenue by Age Group



Key Takeaway

Young Adults lead in revenue contribution at **\$62,143**, with a gradual decline across older groups.

Revenue distribution is relatively balanced — no single age group dominates, suggesting broad market appeal.

- Express shipping users spend slightly more on average (\$60.48 vs. \$58.46 for Standard).

Power BI Dashboard

An interactive dashboard presents all key insights visually with filters for Subscription Status, Gender, Category, and Shipping Type.

3.9K Customers

Total transactions tracked

\$59.76 Avg Purchase

Mean transaction value

3.75 Avg Rating

Customer review score

27% Subscribers

Subscription adoption rate

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to increase the 27% adoption rate.



Loyalty Programs

Reward repeat buyers to convert Returning customers into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on high-discount products like Hats and Sneakers.



Targeted Marketing

Focus campaigns on Young Adults, high-rated products, and express-shipping users.